StockFlow

Inventory management with a stock movement ledger

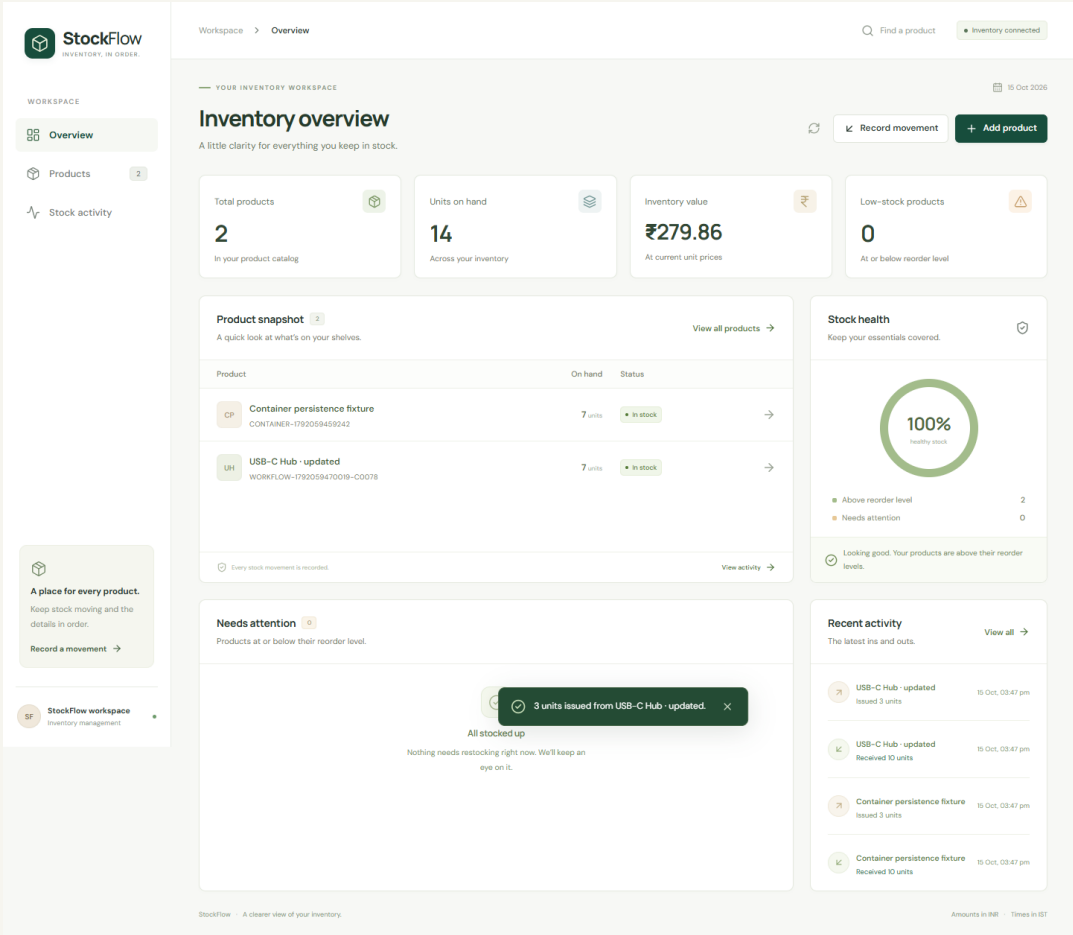
Piyush

DevSecOps capstone | 15 October 2026

Local implementation verified. AWS deployment remains pending.

Inventory workflow

- Product catalog
- Receipts and issues
- Append-only movement history
- Low-stock reporting



Source: StockFlow application and docs/CONTAINERS.md | Actual local screenshot

Application architecture

React / Nginx → **FastAPI** → **PostgreSQL 16**

Browser calls use relative /api paths.

Alembic applies the schema before API readiness.

Row locks serialize competing stock changes.

Inventory and ledger writes share one transaction.

Source: backend/app/, backend/alembic/, docker-compose.yml

Tests and local containers

49 backend tests | 11 frontend unit tests | 8 browser workflows

Real PostgreSQL tests cover concurrency and rollback.

Disposable test databases protect development inventory.

Compose starts the database, migration, API, and UI.

Both application images run as non-root users.

Source: docs/CI_RELEASE.json, docs/FRONTEND_VERIFICATION.md, docs/CONTAINERS.md

CI and image security

Tests > image build > Trivy gate > SHA-tagged GHCR images

HIGH/CRITICAL image findings fail publication.

The pipeline publishes the same scanned image bytes.

Pull requests receive no package publication credentials.

Terraform has three separate documented scan findings.

Source: .github/workflows/ci.yml and docs/CI_SECURITY.md

AWS infrastructure - deployment pending

Sydney | EKS | private worker | NAT | public ALB | gp3

Reviewed plan: 59 additions; one m7i-flex.large worker.

Infrastructure estimate: $0.29/hour.

Illustrative full demo estimate: $0.33/hour.

A single worker has no worker redundancy.

Source: docs/INFRASTRUCTURE_REVIEW.json | USD estimate, before variable usage and credits

Kubernetes and Helm

Two frontend and backend replicas with health probes.

Persistent PostgreSQL PVC and ordered migration Job.

Helm upgrade and database pod replacement retained stock.

Cloud EBS/IRSA and ALB routing still need verification.

Source: docs/HELM.md and docs/HELM_VERIFICATION.json | Local verification

GitOps release and rollback

Green source CI > release-values commit > ArgoCD sync

Initial sync: Synced and Healthy, with migration hooks.

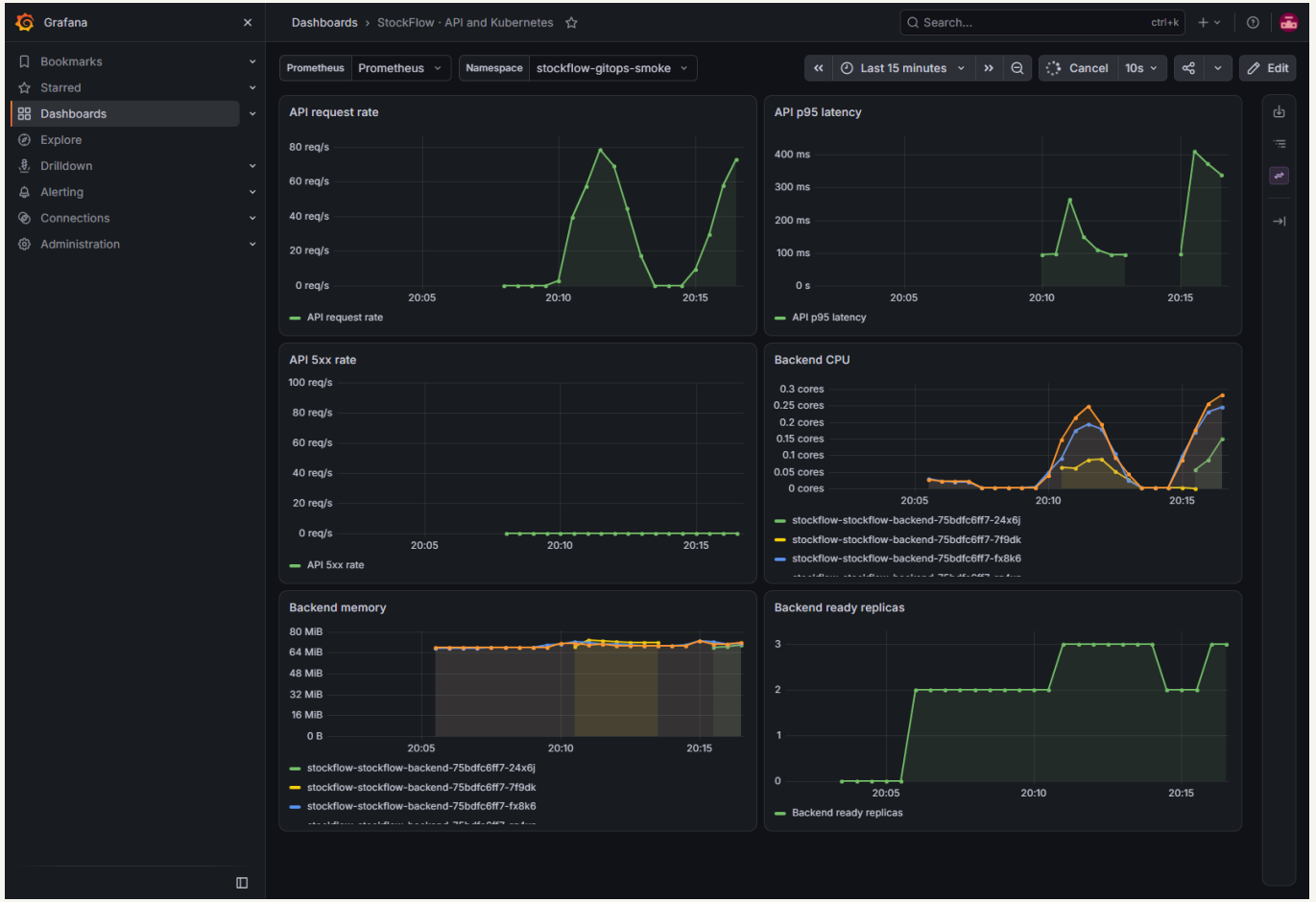
A Git promotion changed both published image SHA tags.

Reverting that commit restored the previous image release.

Quantity 7 and the same PVC survived both changes.

Source: docs/GITOPS_VERIFICATION.json and argocd/README.md | Local ArgoCD

Live monitoring



8,780 read requests; 0 failures; HPA observed 2 > 3 ready replicas

Source: docs/OBSERVABILITY_VERIFICATION.json | Actual local Grafana screenshot

Recovery and lessons

A replacement backend Pod restored healthy replicas.

Persistent inventory survived release and rollback.

ArgoCD sync must preserve database migration hooks.

Longer monitoring exposed Grafana sidecar memory limits.

Source: docs/GITOPS_VERIFICATION.json and docs/OBSERVABILITY_VERIFICATION.json

Evaluation and cleanup

Next: deploy and verify EKS, EBS, public ALB, and cloud GitOps.

Record the mandatory walkthrough with actual cloud evidence.

Keep the live demo until evaluation, unless instructed otherwise.

Back up data, remove the ALB/workloads, then destroy Terraform.

Destroy stops the public URL. Source code and saved evidence remain.

Source: docs/INFRASTRUCTURE.md and demo/RECORDING.md | Cloud evidence remains pending